

Literature Status for the Rigid-Invertibility Questions in 1301.1812

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Status

This is a literature-status packet, not a new proof packet. The two rigid operator questions asked by Costakis, Manoussos and Parissis in [1, Question 2.22, p. 12] are already answered negatively by Mazet and Saias [2, pp. 2–3].

Source Question

The source paper is G. Costakis, A. Manoussos and I. Parissis, *Recurrent Linear Operators*, arXiv:1301.1812. In Question 2.22 on page 12 of the arXiv PDF, after proving that uniformly rigid operators are invertible, the authors ask:

whether every rigid operator is invertible. Failing that, is it true that if T is rigid and invertible then T^{-1} is also rigid?

Question 2.22. We have already seen that if an operator T is invertible then T is recurrent if and only if T^{-1} is recurrent. On the other hand, for uniformly rigid operators we get that T is automatically invertible. However, it is not clear whether T^{-1} is also uniformly rigid without some additional information on T . See also Proposition 3.4. It is natural to ask if the same property is shared by rigid operators, namely, whether every rigid operator is invertible. Failing that, is it true that if T is rigid and invertible then T^{-1} is also rigid? Note that both questions above have affirmative answers in all the examples of rigid and uniformly rigid operators appearing in this paper.

Supporting Answer

Mazet and Saias [2] explicitly identify the Costakis–Manoussos–Parissis questions. Their numbering refers to Question 2.20, while the local arXiv PDF crop above labels the same passage Question 2.22. On page 2 they restate:

Question 1. Are all rigid operators invertible?

Question 2. For every rigid and invertible operator, is its inverse rigid?

They then state that the answer to both questions is negative and prove the following theorem: for every $r \in [0, 1]$, there exists a rigid block-diagonal operator with spectrum

$$\{\lambda \in \mathbb{C} : r \leq |\lambda| \leq 1\}.$$

For $r = 0$ the operator is not invertible, giving a negative answer to the first question. For $0 < r < 1$, the operator is invertible, but the spectrum of its inverse is

$$\{\lambda \in \mathbb{C} : 1 \leq |\lambda| \leq r^{-1}\},$$

so the inverse is not rigid because a rigid operator has spectral radius at most one. Mazet and Saias state this conclusion for the second question on the top of page 3.

celle de [2]).

Enfin quid des opérateurs rigides? La proposition CMP s’étend-elle à ceux-ci? Dans leur question 2.20 ils posent deux questions précises.

Question 1.

Est-ce que tous les opérateurs rigides sont inversibles?

Question 2.

Est-ce que pour tout opérateur rigide et inversible, son inverse est rigide?

La réponse aux deux questions est négative. Nous introduisons au chapitre suivant la notion d’opérateur diagonal par blocs qui étend celle d’opérateur diagonal de $\ell^2(\mathbb{N}^*)$. On a alors le

THÉORÈME.

Pour tout $r \in [0, 1]$, il existe un opérateur diagonal par blocs et rigide dont le spectre est $\{\lambda \in \mathbb{C} : r \leq |\lambda| \leq 1\}$.

Pour $r = 0$, un tel opérateur est non inversible. Cela répond donc négativement à la Question 1. Signalons qu’après avoir prouvé notre théorème, nous avons été informés par Sophie Grivaux que l’existence d’opérateurs rigides dont le spectre est le disque unité fermé peut également être établie à partir du travail [3].

Pour $r \in]0, 1[$, l’inverse d’un tel opérateur n’est pas rigide. En effet son spectre est $\{\lambda \in \mathbb{C} : 1 \leq |\lambda| \leq r^{-1}\}$. Or il est bien connu que le spectre d’un opérateur rigide est inclus dans le disque unité fermé (voir l’Annexe 2 pour

une preuve rapide qui utilise le théorème de Banach–Steinhaus). Cela répond donc négativement à la Question 2.

Scope

This answers both parts of the source question. It does not merely solve a subcase: the source first asks whether all rigid operators are invertible, and Mazet–Saias give a rigid non-invertible operator.

The source’s fallback asks whether an invertible rigid operator has rigid inverse, and Mazet–Saias give invertible rigid operators whose inverses are not rigid.

Search Evidence

The local run indexes were searched for 1301.1812, `rigid invertible`, `non-invertible rigid`, `Recurrent Linear Operators`, and close variants. Web/arXiv searches for the exact source question and for `On the spectrum of rigid operators Mazet Saias` located arXiv:2101.03751. A later contextual paper, Saavedra and Stadlbauer [3, p. 2], also records that Mazet and Saias constructed rigid operators which are either non-invertible or have non-rigid inverse, providing negative answers to two questions of Costakis–Manoussos–Parissis.

Human Review Recommendation

No new mathematical claim is promoted here. Human review should treat the target as closed by the Mazet–Saias literature answer and use this packet to avoid reworking the same source question.

References

- [1] G. Costakis, A. Manoussos and I. Parissis, *Recurrent Linear Operators*, arXiv:1301.1812; Complex Analysis and Operator Theory 8 (2014), 1601–1643.
- [2] P. Mazet and E. Saias, *Sur le spectre des opérateurs rigides*, arXiv:2101.03751; published in English as *On the spectrum of rigid operators*, Ergodic Theory Dynam. Systems 43 (2023), 1351–1362.
- [3] M. Saavedra and M. Stadlbauer, *Quasi-rigid operators and hyper-recurrence*, arXiv:2403.17904.